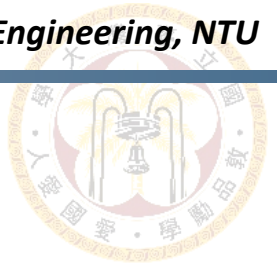
Advanced Computer-Aided VLSI System Design

Homework 1: Lossy Bitmask Encoder with Low Power Design

Graduate Institute of Electronics Engineering, National Taiwan University

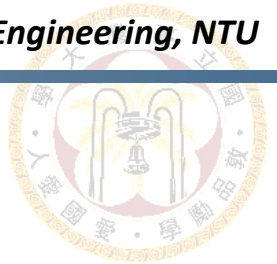


NTU GIEE



Goal

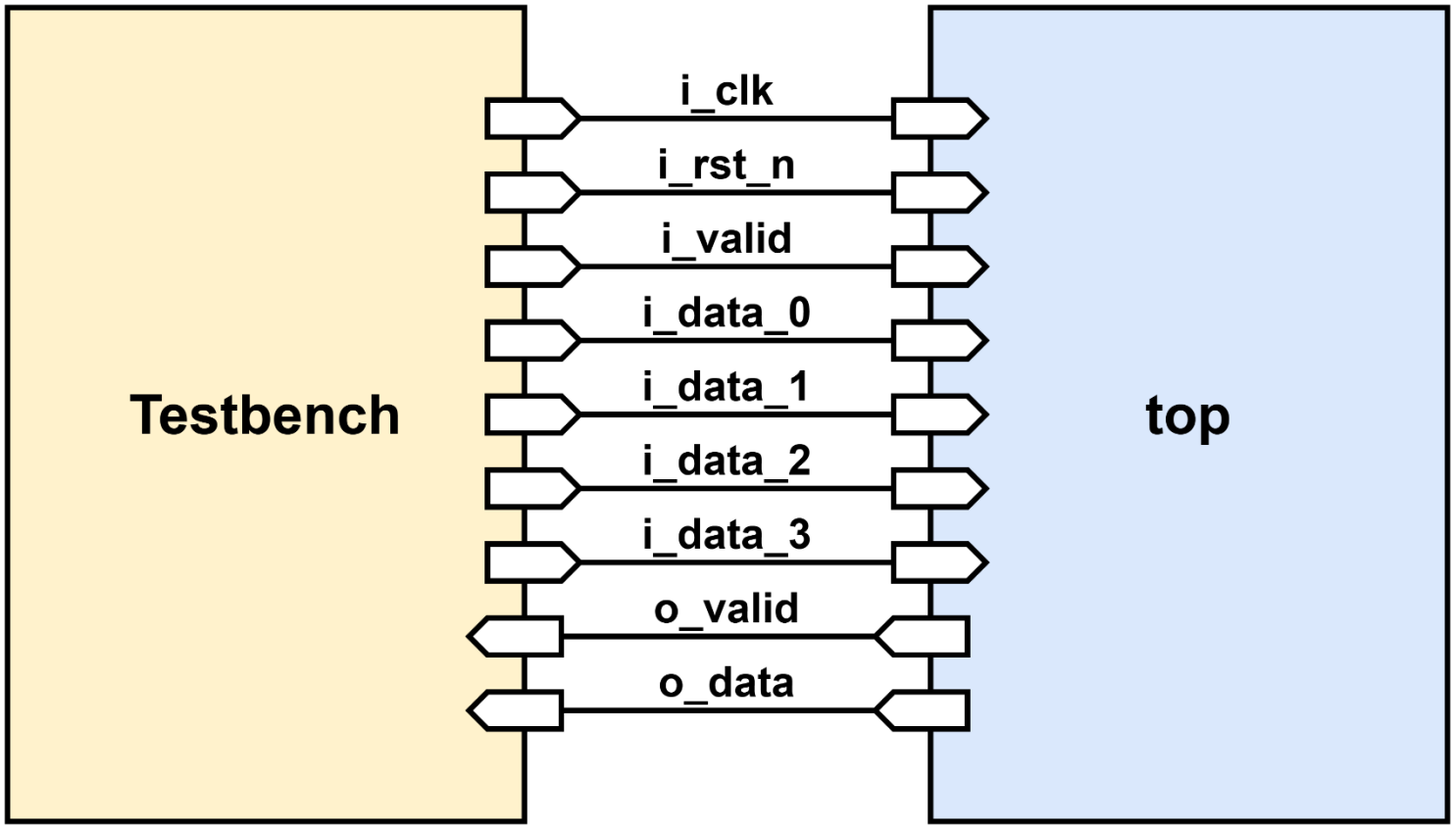
- In this homework, you will learn
  - The flow of the low-power solution
  - Write UPF file to control power domain
  - Implement a low-power design constraint through power reduction methods
  - Simulation commands with VCS NLP
  - Use primetime to estimate power



Introduction

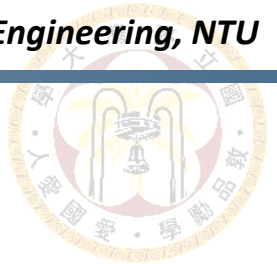
- In this homework, you are going to design a Lossy Bitmask Encoder and implement it with low-power design constraint through UPF flow
- Bitmask encoding is a method for compressing sparse data
- A lossy bitmask encoder treats small values as zeroes, achieving a larger compression ratio at the cost of some error

Block Diagram



Input/Output

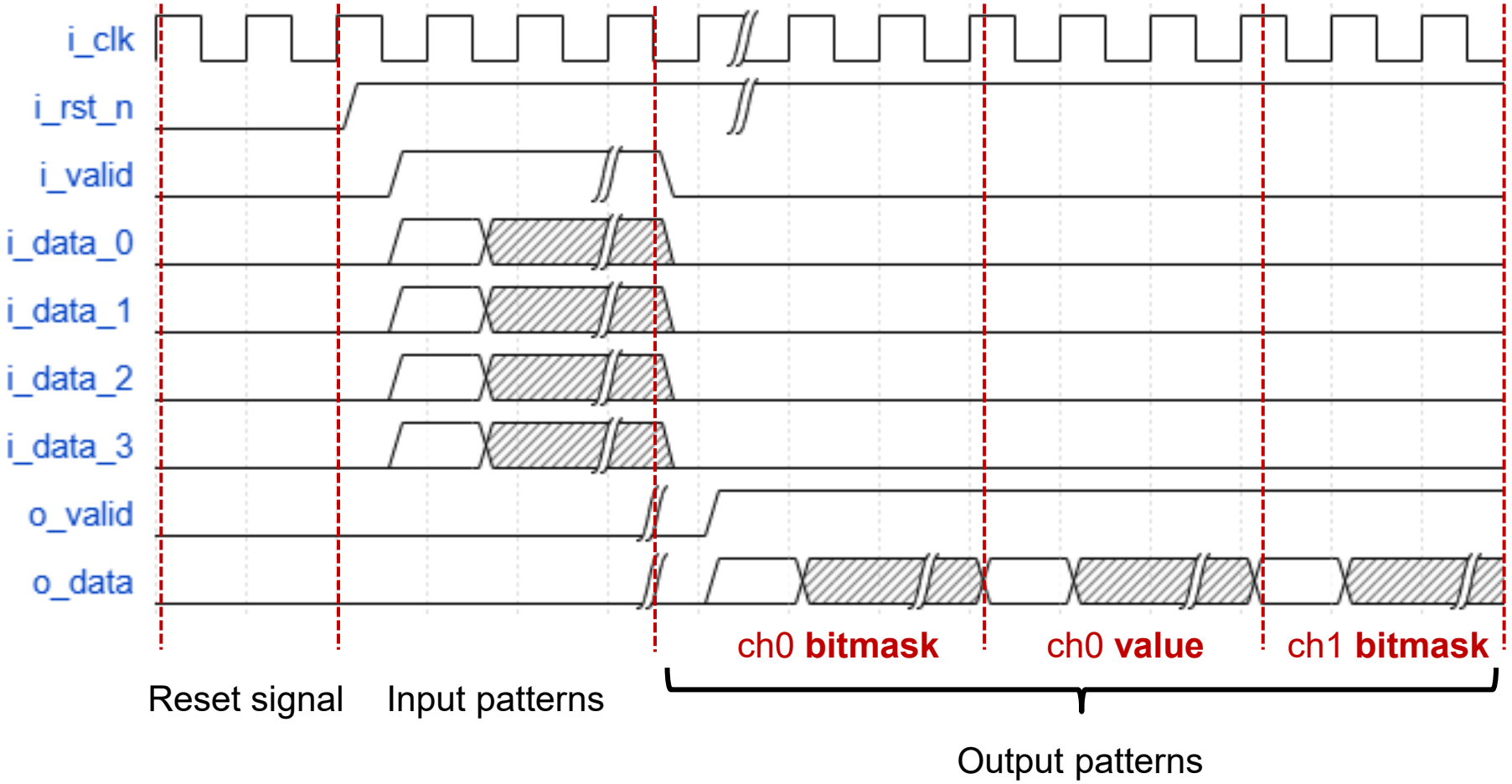
Signal Name	I/O	Width	Simple Description
i_clk	I	1	Clock signal in the system
i_rst_n	I	1	Active **low** asynchronous reset
i_valid	I	1	The signal is **high** if input data is ready
i_data_0	I	8	Unsigned integer input data from 4 channels. Value range 0 ~ 255.
i_data_1	I	8	
i_data_2	I	8	
i_data_3	I	8	
o_valid	O	1	Set **high** if ready to output result, **raise only once**
o_data	O	8	Unsigned output data after bitmask encoding: 1. For the first 1024 cycles, output the bitmask (packed into bytes), then output the real values sequentially 2. Repeat for all 4 channels



Specification (1/2)

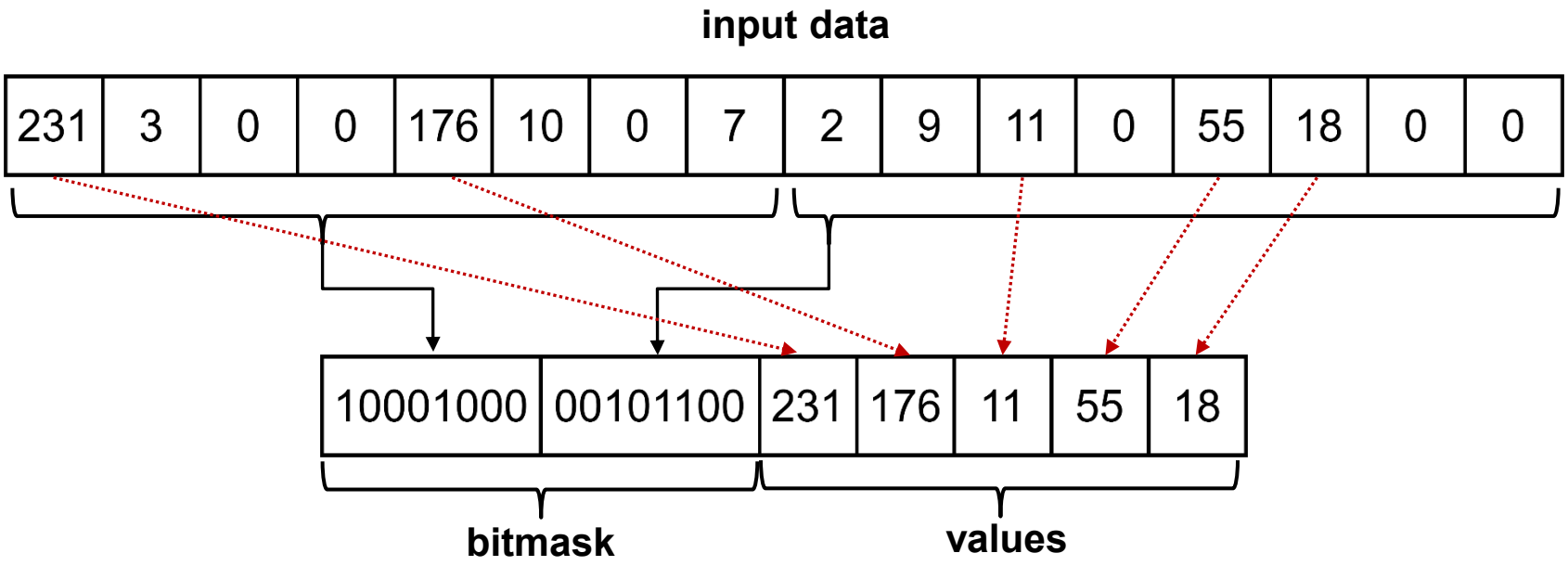
- Active low asynchronous reset is used only once
- All inputs are synchronized with the **negative** clock edge
- All outputs should be synchronized with the **positive** clock edge
- The input pattern (i_data_0, i_data_1 and i_data_2, i_data_3) is ready only when i_valid is high
- o_valid should be high for only one cycle for each o_data
- The testbench will sample o_data at **negative** clock edge if o_valid is high
- You can raise o_valid at any moment **but only once**

Specification (2/2)



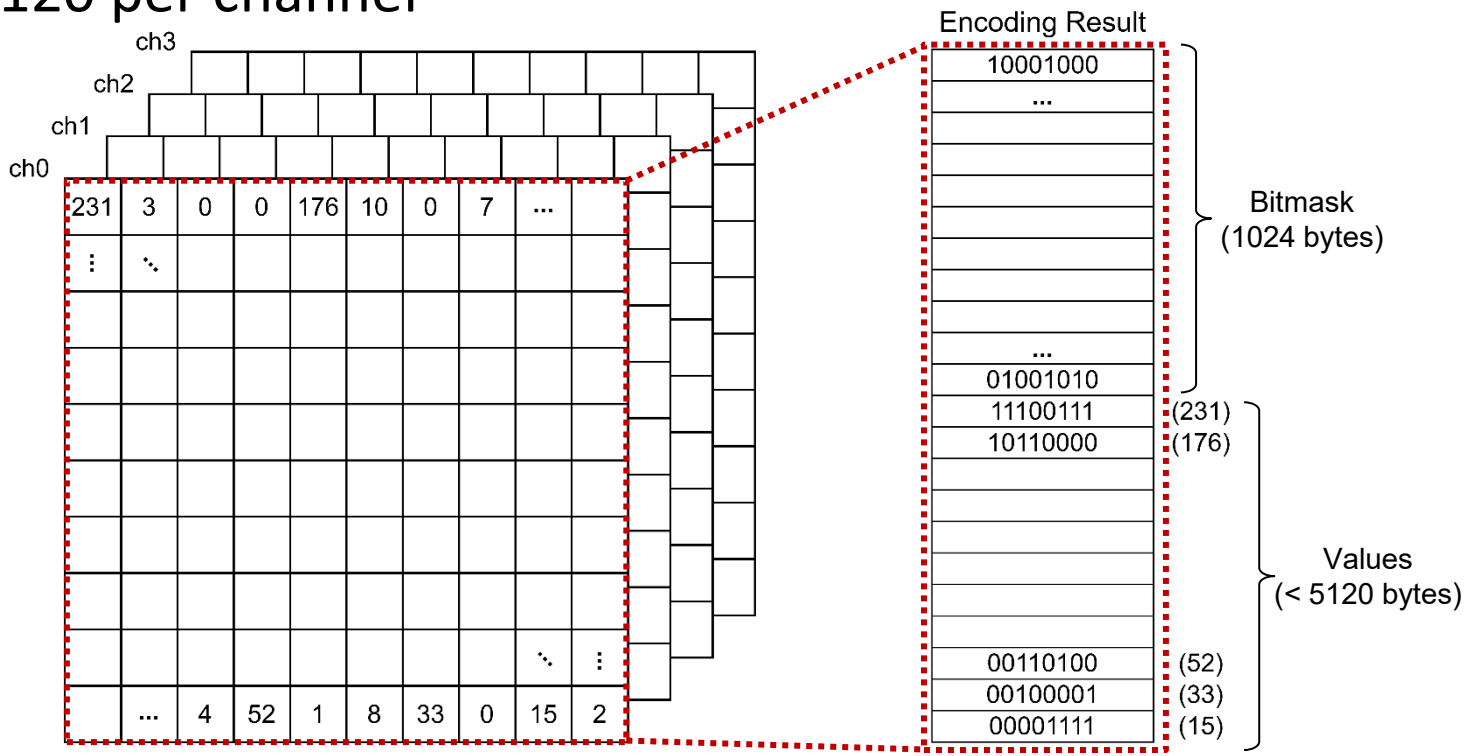
Bitmask Encoding (1/2)

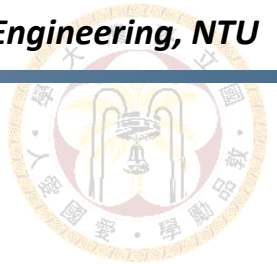
- Lossy Bitmask Encoding:
  - Values lower or equal to the threshold (10) is discarded
  - Bitmask is packed into bytes



Bitmask Encoding (2/2)

- Each channels has 8192 elements
  - Bitmask is 8192/8 = 1024 bytes per channel
  - #Values after compression is guaranteed to be less than 5120 per channel





Requirements for UPF

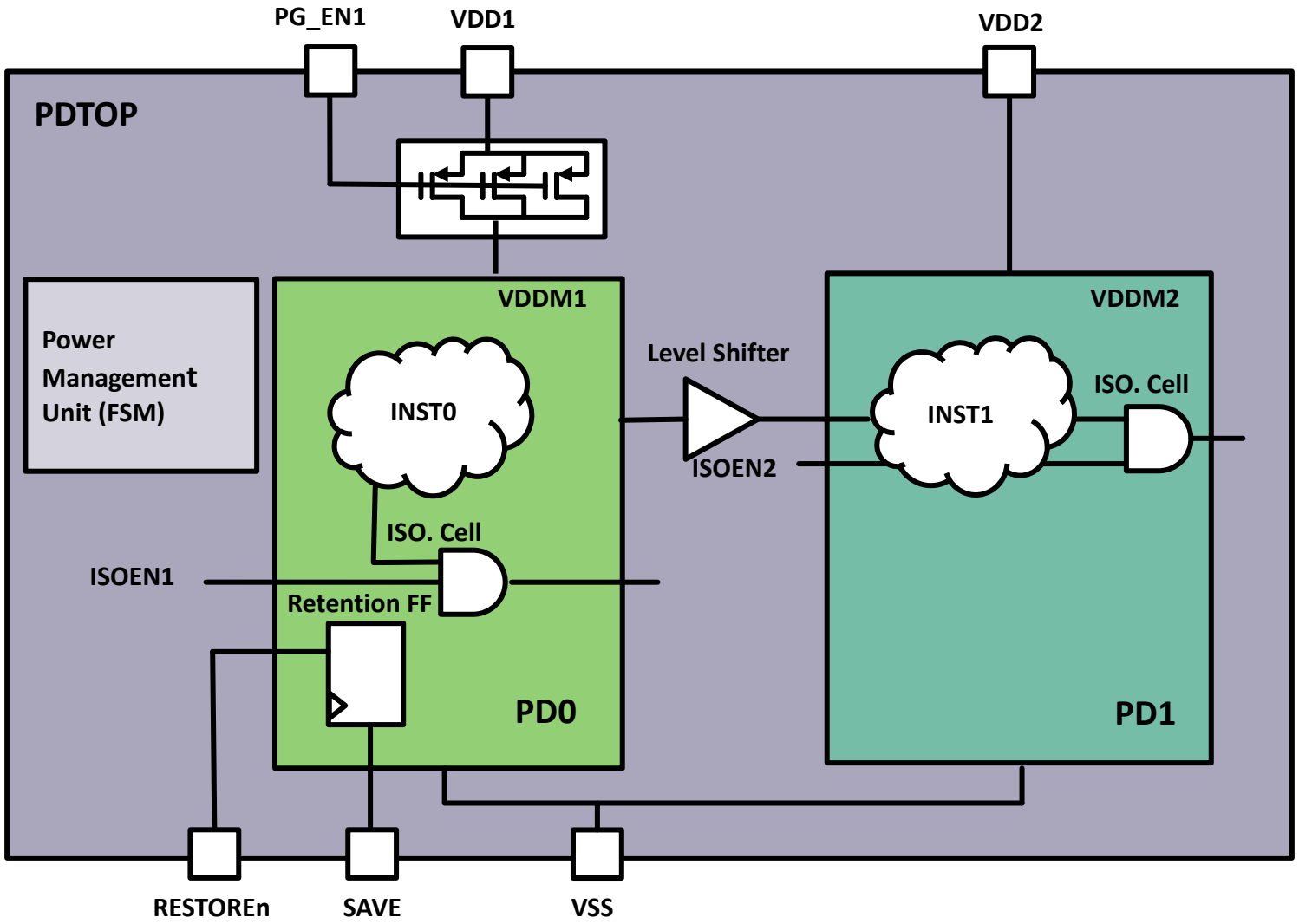
- Power top should be set in the UPF file
- At least **two power domains** (including top domain) should be used in your design
- At least **one power switch** is used in your design

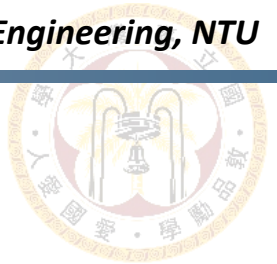
```
set_design_top top
```

```
create_power_domain PD_TOP -include_scope
create_power_domain PD_CORE -elements {u_core}
```

```
create_power_switch psw_core \
    -domain PD_CORE \
    -input_supply_port  {TVDD VDD} \
    -output_supply_port {VDD VVDD_CORE} \
    -control_port       {NSLEEPIN u_core/PSW_NSLEEPIN_CORE} \
    -ack_port           {NSLEEPOUT u_core/PSW_ACK_CORE} \
    -on_state           {state_on TVDD {NSLEEPIN}} \
    -off_state          {state_off {!NSLEEPIN}}
```

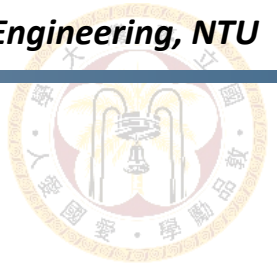
Reference for Low Power Strategy





Other Requirements

- You are **NOT** allowed to use DesignWare
- The range threshold value is set to **10**
- The input length of each channel is **8192**
- **At least one SRAM** should be implemented in your design



top.v

```
module top(
    input i_clk,
    input i_rst_n,
    input i_valid,
    input [7:0] i_data_0,
    input [7:0] i_data_1,
    input [7:0] i_data_2,
    input [7:0] i_data_3,
    output o_valid,
    output [7:0] o_data
);
```

```
);

    // Parameters

    // Wires and Regs

    // Continuous Assignments

    // Sequential Blocks

    // SRAM

    // Combinaional Blocks

endmodule
```

N16 ADFP Memory

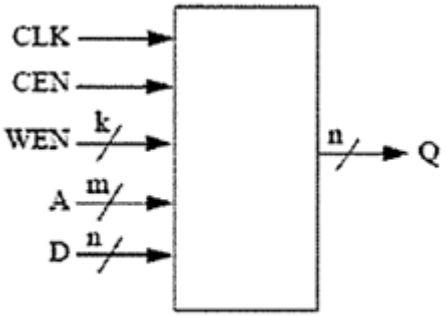
- No memory compiler in N16 ADFP
- Can only use the provided SRAM

TS1 (single port)	TS6 (two port)
TS1N16ADFPCLLLVTA16X88M2SWSHOD	TS6N16ADFPCLLLVTA16X32M2FWSHOD
TS1N16ADFPCLLLVTA16X96M2SWSHOD	TS6N16ADFPCLLLVTA16X72M2FWSHOD
TS1N16ADFPCLLLVTA128X64M4SWSHOD	TS6N16ADFPCLLLVTA16X120M2FWSHOD
TS1N16ADFPCLLLVTA512X45M4SWSHOD	TS6N16ADFPCLLLVTA32X32M2FWSHOD
	TS6N16ADFPCLLLVTA128X32M4FWSHOD
	TS6N16ADFPCLLLVTA128X64M4FWSHOD

Single Port Memory IO

- CEB: active low chip enable
- WEB: write: 1'b0, read: 1'b1
- DSLP/SLP: (Deep) Sleep
- SD: Shut Down

```
module TS1N16ADFPCLLLVTA128X64M4SWSHOD(
    .A       (address),
    .CEB     (cen),  // active low
    .CLK     (clk),
    .WEB     (wen),  // write: low, read: high
    .D       (datain),
    .Q       (dataout),
    .BWEB    ({64{1'b0}}),
    .RTSEL   (2'b01),
    .WTSEL   (2'b01),
    .SLP     (1'b0),
    .DSLP    (1'b0),
    .SD      (1'b0),
    .PUDELAY ()
);
```

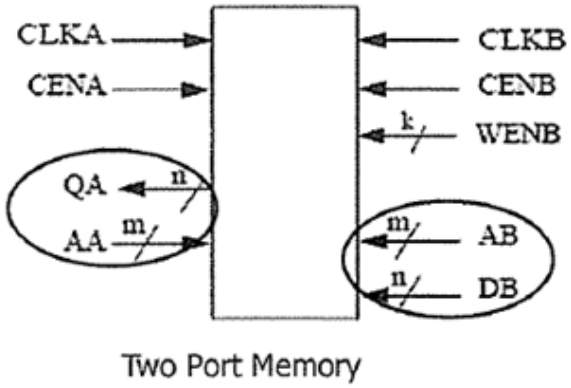


Single Port Memory

Two Port Memory IO

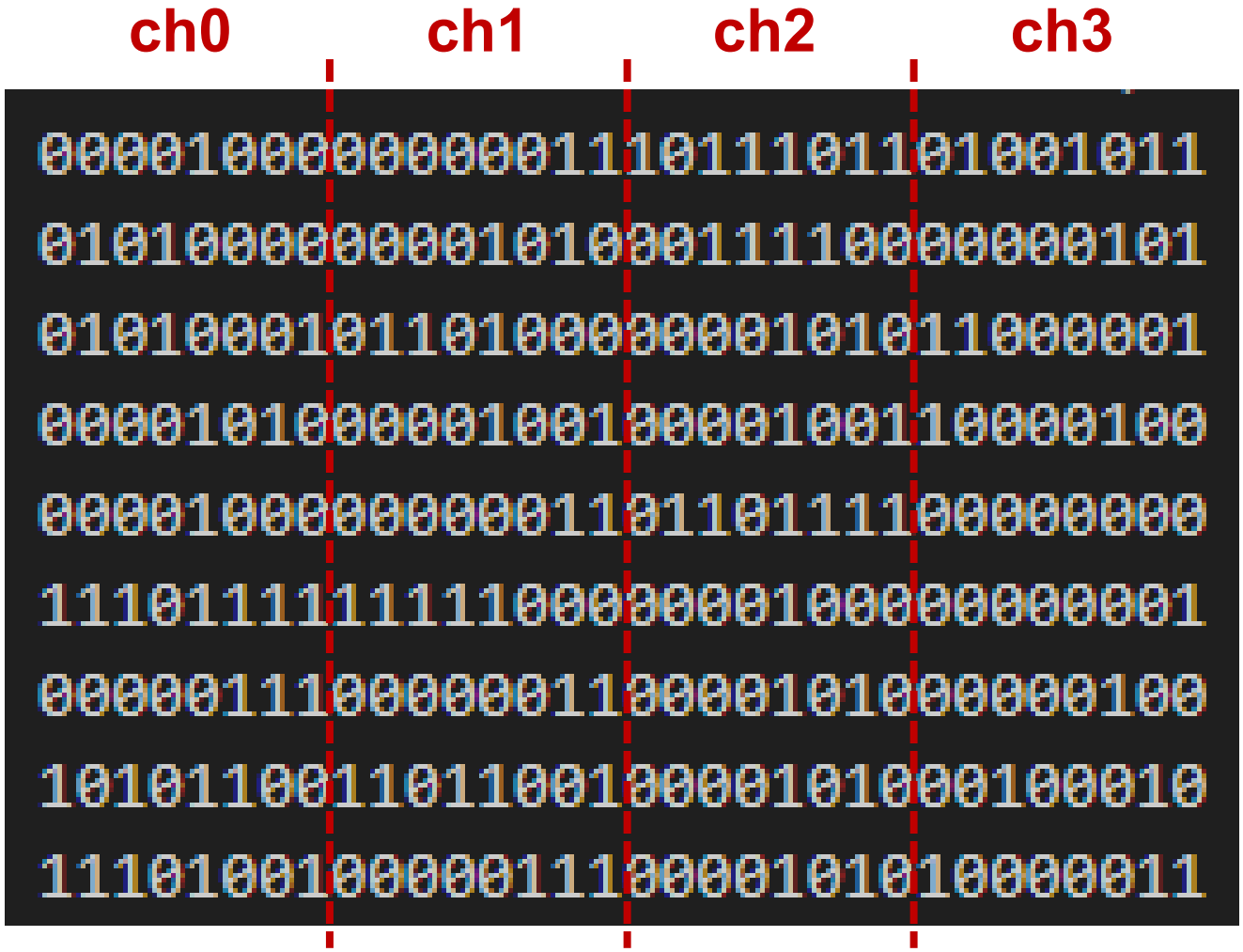
- REB: active low read enable
- WEB: active low write enable

```
module TS6N16ADFPCLLLVTA128X32M4FWSHOD(
    .AB       (read_address),
    .REB      (read_en),    // active low
    .CLKR     (read_clk),
    .Q        (dataout),
    .AA       (write_address),
    .WEB      (write_en),   // active low
    .D        (datain),
    .CLKW     (write_clk),
    .BWEB     ({32{1'b0}}),
    .RCT      (2'b01),
    .WCT      (2'b01),
    .KP       (3'b011),
    .SLP      (1'b0),
    .DSLP     (1'b0),
    .SD       (1'b0),
    .PUDELAY ()
);
```



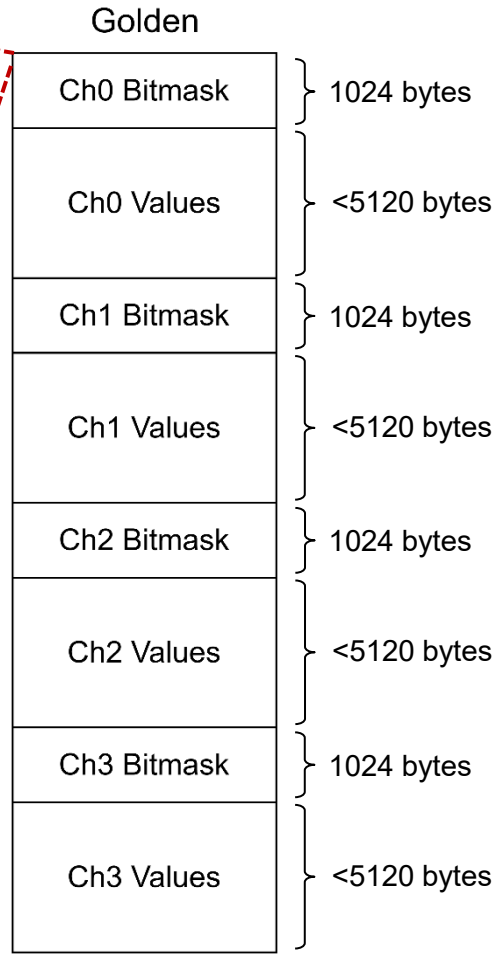
Two Port Memory

Pattern (Input Data)



Pattern (Golden Output)

```
1    01100101
2    10100100
3    01111101
4    01010001
5    00011111
6    00101100
7    00001000
8    10000010
9    00101010
10   11100000
11   00010000
12   01010100
13   00101011
14   00000101
15   10101011
16   10110100
```



Submission

- Create a folder named **studentID_hw1** and follow the hierarchy below (*.sv is allowed if you use SystemVerilog)

```
r14943009_hw1
├── 01_RTL
│   ├── top.v
│   ├── xxx.v (other verilog files)
│   └── rtl_01.f
├── 02_GATE
│   ├── top_syn.v
│   ├── top_syn.sdf
│   └── rtl_02.f
├── 03_SYN
│   ├── top_syn.area
│   ├── top_syn.timing
│   ├── top_syn.ddc
│   └── top_syn.tcl
├── 04_UPF
│   ├── top.rtl.upf
│   └── top.syn.upf
├── 05_POWER
│   ├── p0.power
│   ├── ...
│   └── p4.power
└── report.txt
```

Submission - Workstation

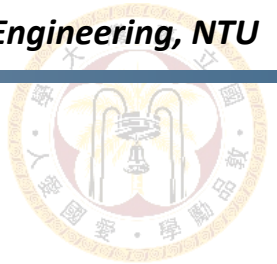
- Pack the folder **studentID_hw1** into a **tar.gz file** named **acvsd-xx-hw1.tar.gz**
  - **tar -zcvf acvsd-xx-hw1.tar.gz [path to studentID_hw1]**
  - Use **lowercase** for all the letters (e.g. acvsd-11-hw1.tar.gz)
  - Pack the folder on ADFP server to avoid OS related problems
  - Place the tar.gz file at the home of your ADFP account

- TA will only check the last version after the homework deadline
- Reminder
  - Files uploaded to the ADFP workstation must be kept as a copy in your local folder or computer

Submission - NTU COOL

- For design files and UPF scripts (01_RTL, 04_UPF, report)
- Pack the folder **studentID_hw1** into a **tar.gz file** named **acvsd-xx-hw1-vk.tar.gz** (*k* is the number of version, *k* =1,2,…)
  - **tar -zcvf acvsd-xx-hw1-vk.tar.gz [path to studentID_hw1]**
  - Use **lowercase** for all the letters. (e.g. acvsd-11-hw1-v1.tar.gz)
  - Pack the folder on ADFP server to avoid OS related problems
- Submit to NTU Cool

```
NTU Cool
r14943009_hw1
├── 01_RTL
│   ├── top.v
│   ├── xxx.v (other verilog files)
│   └── rtl_01.f
├── 04_UPF
│   └── top.rtl.upf
└── report.txt
```



Grading Policy (1/3)

- Grading command for RTL
  - `vcs -full64 -R -f rtl_01.f +v2k -sverilog -debug_access+all +define+$1`
  - `vcs -full64 -R -f rtl_01.f –upf [path to UPF file] +v2k -sverilog -debug_access+all +define+$1+UPF`
- Simulation: **70%**
  - Need to pass all of 5 public and 5 private test cases

	Score
RTL simulation	15%
RTL with UPF	15%
Gate-level simulation	20%
Gate-level with UPF	20%

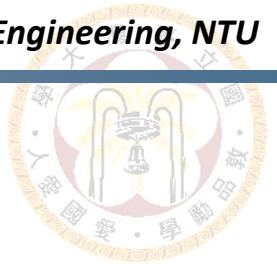
Grading Policy (2/3)

- Grading command for gate-level
  - `vcs -full64 -R -f rtl_02.f +v2k -sverilog -v2005 -debug_access+all +maxdelays -negdelay +neg_tchk +define+$1+SDF`
  - `vcs -full64 -R -f rtl_02.f –upf [path to UPF file] +v2k -sverilog -v2005 -debug_access+all +maxdelays -negdelay +neg_tchk +define+$1+SDF+UPF`
- Performance: **30%**
  - Use public test cases (But need to pass hidden pattern)
  - Performance = $\sum(P^2 \times T) \times A$
    - P: Power (mW), T: Simulation time (ns), A: Area (um^2)
  - Baseline = 1×10^7

	Score
Baseline	10%
Ranking (Need to pass baseline)	20%

Grading Policy (3/3)

- Lose 5 points for any incorrect naming or format
- Violations of any spec incur point penalties
  - Negative slack/Design without SRAM
    - 0 point for gate-level simulations and performance
  - Violate other rules but pass all simulations
    - Performance score * 0.7
- No late submission
  - **0 point** for this homework
- No plagiarism
  - Plagiarism in any form, including copying from online sources, is strictly prohibited



Time

- Time: Processing time from simulation (e.g. 37814.25ns below)

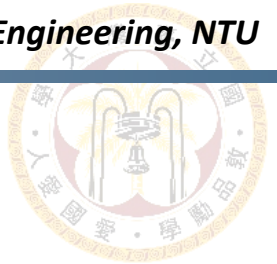
```
-----------------------------------------------
-                 ALL PASS!                   -
-----------------------------------------------
$finish called from file "../00_TESTBED/testbench.v", line 239.

*************Low Power Message Summary*******************

INFO = 40, WARNING = 0, ERROR = 0, FATAL = 0

*****************************************************

$finish at simulation time              37814250
           V C S   S i m u l a t i o n   R e p o r t
Time: 37814250 ps
```



Area

- Area: Total cell area from synthesis report (e.g. 53123.3 μm^2 below)

```
Number of ports:                          647
Number of nets:                          5374
Number of cells:                         4156
Number of combinational cells:           3445
Number of sequential cells:               696
Number of macros/black boxes:              12
Number of buf/inv:                       1198
Number of references:                      52

Combinational area:               1176.145948
Buf/Inv area:                      311.402882
Noncombinational area:             801.861150
Macro/Black Box area:            51145.291016
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                 53123.298114
Total area:                 undefined
```

Number of macros/black boxes should not be 0

Power

- Use Primetime to calculate power (e.g. 0.0194mW below)

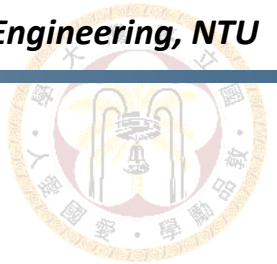
```
                          Internal  Switching  Leakage    Total
Power Group               Power     Power      Power      Power   (     %)  Attrs
--------------------------------------------------------------------------------
clock_network             1.065e-03    0.0000    0.0000 1.065e-03 ( 5.50%)  i
register                  5.038e-05 4.722e-05 1.227e-08 9.761e-05 ( 0.50%)
combinational             1.232e-04 7.289e-05 2.799e-07 1.964e-04 ( 1.01%)
sequential                   0.0000    0.0000    0.0000    0.0000 ( 0.00%)
memory                       0.0180 1.757e-06 3.806e-06    0.0180 (92.98%)
io_pad                       0.0000    0.0000    0.0000    0.0000 ( 0.00%)
black_box                    0.0000    0.0000    0.0000    0.0000 ( 0.00%)

  Net Switching Power  = 1.219e-04   ( 0.63%)
  Cell Internal Power  =    0.0192   (99.35%)
  Cell Leakage Power   = 4.099e-06   ( 0.02%)
                         ---------
Total Power            =    0.0194  (100.00%)

X Transition Power     = 1.471e-07
Glitching Power        = 3.371e-07

Peak Power             =    5.2047
Peak Time              =  6687.000
```

Power = 0.0194 mW



Report

```
StudentID: r14943009
Clock period: 1.5 (ns)
Area: 53123.30 (um^2)
-------------------------
p0 time: 37814.25(ns)
p0 power: 0.0194 (mW)
-------------------------
p1 time: 40628.25 (ns)
p1 power: 0.0190 (mW)
-------------------------
p2 time: 43227.75 (ns)
p2 power: 0.0186 (mW)
-------------------------
p3 time: 45170.25 (ns)
p3 power: 0.0184 (mW)
-------------------------
p4 time: 47858.25 (ns)
p4 power: 0.0180 (mW)
-------------------------
```

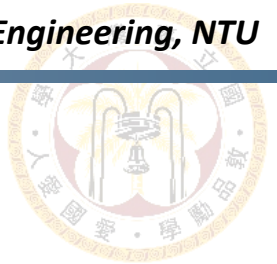
Discussion

- **NTU Cool Discussion Forum**
  - For any questions not related to assignment answers or privacy concerns, please use the NTU Cool discussion forum.
  - **TAs will prioritize answering questions on the NTU Cool discussion forum**

- **Email:** r14943009@ntu.edu.tw
  - Title should start with **[ACVSD 2026 Spring HW1]**
  - Email with wrong title will be moved to trash automatically

Discussion


課程內容
課程資訊
公告
作業
討論
Gradescope
成績
設定
[HW1]Discussion
所有班別
HW1相關問題在此討論，並請以下列格式發問，方便助教按照每個問題回答
1. 問題一
2. 問題二
...
另外，若需要截圖，請勿把自己的code截圖或code文字上傳，變成大家的參考答案，若違反將扣本次作業總分10分。
祝同學們學習順心
by TA
[提醒]
1. ...
2. ...
3. ...



References

- [1] Reference for bitmask encoding (If there is any conflict, the homework instructions shall prevail)
  - Bitmask Encoding
- [2] Reference for UPF
  - Synopsys ® Multivoltage Flow User Guide